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Comparison of fatigue accumulated during and after prolonged robotic and laparoscopic surgical methods: a cross-sectional study

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Abstract
Background The aim of the present study was to analyse the fatigue experienced by surgeons during and after performing robotic and laparoscopic surgery and to analyse muscle function, self-perceived fatigue and postural balance.
Methods Cross-sectional study considering two surgical protocols (laparoscopic and robotic) with two different roles (chief and assistant surgeon). Fatigue was recorded in two ways: pre- and post-surgery using questionnaires [Profile of Mood States (POMS), Quick Questionnaire Piper Fatigue Scale and Visual Analogue Scale (VAS)-related fatigue] and parametrising functional tests [handgrip and single-leg balance test (SLBT)] and during the intervention by measuring the muscle activation of eight different muscles via surface electromyography and kinematic measurement (using inertial sensors). Each surgery profile intervention (robotic/laparoscopy–chief/assistant surgeon) was measured three times, totalling 12 measured surgery interventions. The minimal duration of surgery was 180 min.
Results Pre- and post-surgery, all questionnaires showed that the magnitude of change was higher for the chief surgeon compared with the assistant surgeon, with differences of between 10 % POMS and 16.25 % VAS (robotic protocol) and between 3.1 % POMS and 12.5 % VAS (laparoscopic protocol). In the inter-profile comparison, the chief surgeon (robotic protocol) showed a lower balance capacity during the SLBT after surgery. During the intervention, the kinematic variables showed significant differences between the chief and assistant surgeon in the robotic protocol, but not in the laparoscopic protocol. Regarding muscle activation, there was not enough muscle activity to generate fatigue.
Conclusion Prolonged surgery increased fatigue in the surgeon; however, the magnitude of fatigue differed between surgical profiles. The surgeon who experienced the greatest fatigue was the chief surgeon in the robotic protocol.

Keywords Surface EMG · Kinematic measurement · Handgrip · Balance · Profile of Mood States · Quick Questionnaire Piper Fatigue Scale

Minimally invasive surgery offers a number of advantages to the patient, which has encouraged its widespread use despite the complex technical difficulties involved [1]. In addition to these technical difficulties, it has been observed that surgeons operating regularly have adverse symptomatologies due to ergonomic requirements [1]. Among the most common causes of reduced ergonomics during laparoscopy are excessive strain on the neck, held

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abduction of the shoulder, prolonged standing with little change of position and excessive flexing of the hand [2, 3]. This accumulation of malposition negatively affects the postural health of surgeons who perform laparoscopy and causes the appearance of paraesthesia and pain in the hand, wrist, arm, shoulder and neck [2–4].

The execution of surgery in a sitting position, which offers minimal resistance to the movement of the instruments, or three-dimensional visualisation through a console [5], enables the use of robotic surgery and reduces physiological stress for surgeons while maintaining the benefits of minimally invasive laparoscopic surgery [6, 7]. However, the lack of ergonomics could cause the early onset of fatigue. Muscle fatigue could occur during or after any gesture or task that reduces the maximum capacity to generate power or force [8], defined as a state that temporarily limits the ability to perform a task or job precisely because of the same task or job [9]. In addition, when the postural muscles that control balance are fatigued, they may alter their function, negatively conditioning balance and postural stability [10–12]. Some studies analysed the effects of fatigue on the static balance of the individual and concluded that there was a direct correlation between postural muscle fatigue and the static balance of the person [13, 14]. Other studies suggested movement as a strategy to slow the onset of fatigue, especially when space for work was reduced [15, 16].

The detection and classification of neuromuscular fatigue have contributed significantly to information regarding human–computer interactions, sports injuries, performance and ergonomics [17].

Surface electromyography (sEMG) is frequently used to analyse muscle function during the execution of a particular task by a particular muscle and facilitates the analysis of the responses given by a muscle during the execution of a particular gesture [18]. In addition, inertial sensors have frequently been used to analyse the postural balance of individuals [19].

A recent study considered the fatigue accumulated by surgeons when performing an intervention by comparing muscle function and postural stability before and after robotic and laparoscopic surgery [20]. However, no studies have compared the onset of fatigue in the surgeon during robotic and laparoscopic surgery.

The aim of this study was to analyse the fatigue experienced by a surgeon during and after surgery by comparing robotic and laparoscopic surgery and analysing muscle function, self-perceived fatigue and postural balance. It was hypothesised that there would be significant differences during and after surgery between the two methodologies analysed.

Materials and methods

Design

This was a cross-sectional study in which muscle fatigue and postural balance before, during and after two surgical techniques (laparoscopy and robotic) were recorded in a surgeon during surgery. Measurements were taken during digestive surgery with a previously estimated duration of 180 min.

Settings

Measurements were taken in two public hospitals in Malaga between March 2014 and June 2015.

Participants

The subject analysed in this study was a general and digestive surgeon with over 15 years of experience in regularly performing laparoscopic and robotic surgery.

Ethical considerations

The study was conducted according to ethical principles of the Declaration of Helsinki. Ethical approval was received from a local ethics committee.

The principles of confidentiality and autonomy were upheld through informed consent and the segregation of data to ensure anonymity at all times. The data were processed and guarded with respect for the privacy of the subject and the current data protection legislation (Law 15/1999) on the protection of personal data.

Instruments for measuring fatigue

Surgeons' Patient-Reported Outcome Measures (PROMS)

To analyse self-perceived fatigue, the Profile of Mood States (POMS) [21] and the Quick Questionnaire Piper Fatigue Scale (QPFS) [22] were used. The POMS comprises 65 items [item scale 0 (Not at all)–4 (Extremely)], with scores grouped into seven subscales: Anger–Hostility (11 items), Depression–Dejection (14 items), Fatigue–Inertia (8 items) Vigour–Activity (6 items), Friendliness (6 items), Tension–Anxiety (3 items) and Confusion (3 items) [23]. The Vigour subscale had an inverse relationship with the other subscales [24]. This questionnaire presented a reliability (Cronbach's alpha) ranging between 0.77 and 0.92 [21]. The QPFS is a questionnaire with 15 items [response scale 0 (best situation)–10 (worst situation)] based on the Piper Fatigue Scale-Revised and has

advantages such as brevity and the ease of interpretation of results (the results are accumulated in one dimension) [22]. It had a reliability score of 0.947 and a validity of 0.743 [22]. Finally, the Visual Analogue Scale (VAS)-related fatigue was used. The surgeon was asked: What is your level of fatigue after performing surgery? Self-administered questionnaires were completed before and after surgery.

Objective Clinical Outcome Measure (OCOM)

Surface EMG

The electromyography Biomonitor ME6000 (Mega Electronics Ltd.) was used to record muscle activation. The sampling frequency was 2000 Hz. On the dominant side, the following eight muscles were analysed: tibialis anterior, medial gastrocnemius, vastus medialis, biceps muscle femoris, abdominal transverse, erector spinae, medial deltoids and trapezius muscle (Fig. 1). For each muscle, three electrodes were used [Al/AgCl (2.5 cm^2)] with a 1-cm gap between the three. The electrodes were positioned following the recommendations of the European project "Surface Electromyography for the Non-Invasive Assessment of Muscles" (SENIAM) [25, 26] and the electromyography manufacturer (Mega Electronics Ltd) [27]. Before positioning each electrode, each muscle was palpated. In addition, to minimise the electromyographic limitations (impedance of the skin), the skin was shaved and cleaned with alcohol, as recommended by SENIAM.

Electromyographic data collection was conducted using the software Megawin 3.0.1 (Mega Electronics Ltd, Kuopio, Finland). In addition, to remove high-frequency noise, a high-pass corner filter (20 Hz) was used. The maximum voluntary contraction (MVC) of each muscle was obtained according to previously used protocols [28]. Each maximal activation test was performed three times (maximum contraction for 5 s with 30 s of rest between contractions), using, as a reference for normalisation, the maximum activation value registered.

The electromyographic recording was performed during the development of the surgical interventions.

Inertial sensors

An inertial sensor with a sampling frequency of 180 Hz (InertiaCube3TM of InterSense Inc. (Bedford, Massachusetts, USA) was used to collect the surgeon's kinematic variables during surgery. After cleaning the skin, the inertial sensors were fixed to the skin with tape and reinforced with a rubber band. The inertial sensor was placed on the back at the T7 level. Before placing the inertial sensors, they were calibrated positioning the coordinates' centre at the down left corner.

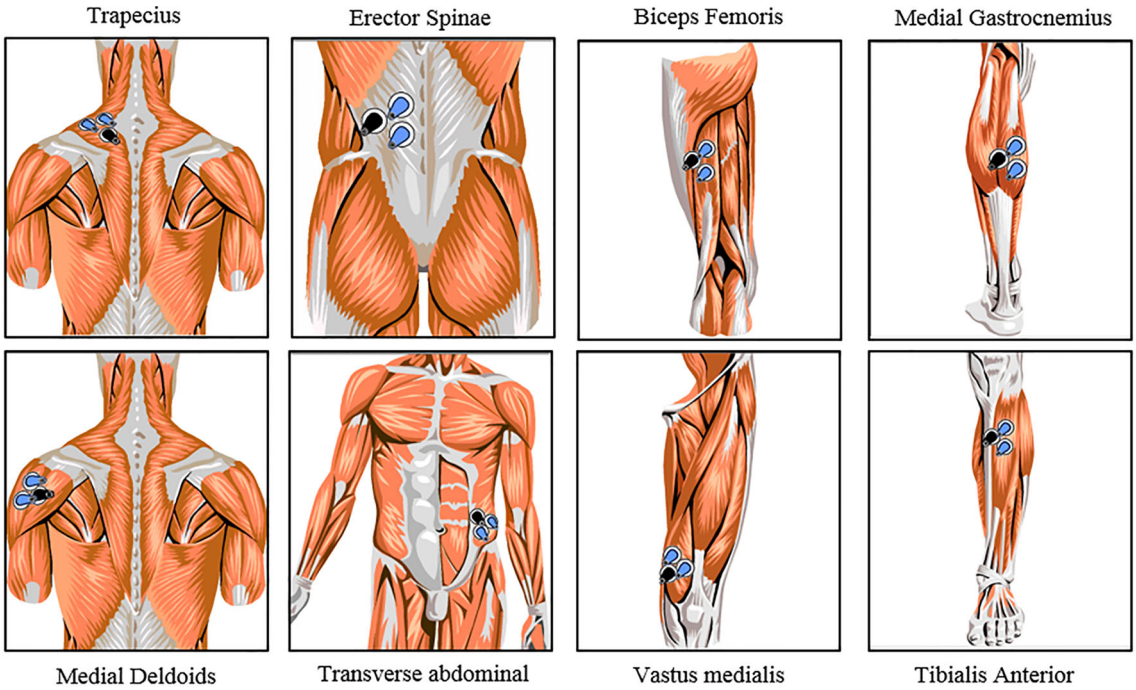

Fig. 1 Positioning scheme of the electrodes according to the analysed muscle. Images taken from the programMegaWin 3.0.1 (Mega Electronics Ltd, Kuopio, Finland)

The inertial sensors created a kinematic record during the single-leg balance test (SLBT) balancing test and during the course of surgery.

Functional tests

The participant performed the SLBT, which was intended to measure static balance. In this test, the participant had to stand on one leg with the other leg in suspension, and the hip in a neutral position with the knee flexed to 90° with his/her hands on the hips. The test was completed when the participant removed the hands from the hips or when the suspended leg touched the ground. If this did not occur, the test ended after the participant had remained static for 30 s. The test was performed in four different settings: supported by dominant and non-dominant legs and with eyes open and closed. In addition, this test was monitored kinematically using the inertial sensors positioned on the back at the T7 level.

Finally, the hydraulic Jamar® handgrip dynamometer (Sammons Preston Rolyan, Chicago, IL) was used three times to perform the maximal handgrip test on both hands (dominant and non-dominant) for 5 s each, with a break of 60 s between repetitions.

The two functional tests were performed before and after each surgery.

Surgical protocols

Two surgical protocols, laparoscopic and robotic surgery using the Da Vinci® Surgical System (Intuitive Surgical—Sunnyvale, CA), were performed in the present study. During the two protocols, the surgeon performed two distinct roles: *chief surgeon*, ordered to perform surgery and make decisions during the operation, and *assistant surgeon*, responsible for assisting the chief surgeon during surgery. The chief surgeon was placed in the control device during the intervention involving robotics, while, in laparoscopic surgery, the surgeon was responsible for performing surgery. The assistant surgeon was responsible for modifying the various arms of the surgical robot in the robotic intervention, whereas in laparoscopic surgery, the assistant surgeon was responsible for holding the camera for the chief surgeon.

Variables

Before beginning the study, the surgeon's anthropometric data were collected. Outcomes were recorded using the Surgeons' Patient-Reported Outcome Measures (PROMS) and the Profile of Mood States (POMS); the main result (the POMS value) and seven sub-scales were calculated based on the number of items. The Quick Piper Fatigue Scale (QPFS) was also used, resulting in a variable obtained from the sum of the 15 items of the questionnaire; the VAS recorded visual analogue scale-related fatigue. The OCOM included the handgrip test [maximum and maximum peak handgrip strength (average of three attempts)] and the SLBT, which measured the time until the first attempt at maintaining balance or when the hands were separated from the hips. The SLBT had a ceiling of 30 s. Kinematic variables recorded during the SLBT and during surgery included (for all axes): mean displacement, mean velocity and time. Electromyography variables were recorded during surgery in all above-mentioned muscles and included median muscle activation frequency (average muscle activation recorded throughout surgery, where mean values presented normalised frequencies) and median muscle activation slope (MDFslope), where the initial and final values of muscle activation were used in a linear regression in which the slope of the linear regression was the fatigue level. Kinematic variables included mean angular displacement (average angular displacement in the three axes recorded during surgery) and average speed (the average speed of displacement during the operation).

Statistical analysis

A descriptive analysis of the anthropometric variables was performed. For each of the four intervention profiles (robotic/laparoscopy–chief/assistant surgeon), along with the outcome variables, a descriptive study of those that were measured during surgery was performed. In addition to the outcome variables measured before and after surgery, based on the distribution of the sample (calculated through the KS test), a pre- and post-intervention comparison was made using Student's *t* test for parametric variables and Wilcoxon's test for nonparametric variables.

In addition, an "inter-profile" comparison comparing all outcome variables between the four intervention profiles was performed. A one-way ANOVA for each of the pre-intervention and post-intervention variables was performed. The level of significance was established as $p \leq 0.05$. For statistical analysis, The Statistical Package for Social Sciences (SPSS) was used (version 21.0 for Windows, Chicago, IL).

Results

Descriptive results

The digestive surgeon was 44 years old, 1.87 cm tall and weighed 92 kg with a BMI of 27.45 kg/m^2. The total number of intervention measures was 12 (three for each of the surgical protocols). There were no significant

differences in the duration of the surgical procedures performed, with an average of 218.46 (±14.55) min. Figure 2 shows the MVC of each muscle recorded during Daniels' test. To normalise the electromyographic measures, the maximal muscle activation recorded during Daniels's test was used. There were no significant differences between surgeon profiles in this study.

Variables measured pre- and post-surgical intervention

Intra-profile differences

Table 1 shows pre- and post-intervention results and the differences between each of the variables during robotic interventions. The results showed significant changes between pre- and post-surgery intervention in all Surgeons' PROMS variables analysed. However, the magnitude of change was much larger for the chief surgeon compared to the assistant surgeon, with a difference equal to or greater than 10 % (10 % POMS; 11.7 % QPFS; 16.25 % VAS) in the two robotic surgery profiles. Significant changes were also observed in the pre- and post-intervention OCOM variables. In the inter-profile comparison, the tests showed that the chief surgeon had a lower balance capacity, showing a significantly greater magnitude of change in all kinematic variables recorded during the SLBT in different conditions (Table 1), with the exception of some speed measurements.

Similarly, when comparing the change in intra-profiles that occurred during laparoscopic intervention, significant differences were observed in all Surgeons' PROMS and OCOM variables, with the exception of velocities measured during the SLBT in both the chief and assistant surgeon profiles (Table 2).

Intra-profile differences after different surgical protocols

Comparison of the variables of the chief surgeon during robotic and laparoscopic surgery revealed that the magnitude of the change experienced by Surgeons' PROMS variables after the robotic protocol with respect to the laparoscopy protocol was significantly higher for all Surgeons' PROMS except $POMS_{Confusion}$ and QPFS. In addition, during the SLBT, the results showed that there was higher functional fatigue in the chief surgeon during robotic surgery. Table 1 shows that there was a significantly greater change in 15 of the kinematic variables following robotic surgery (Table 1), whereas only five of the kinematic variables showed a significantly greater change following laparoscopic surgery (Table 2). Meanwhile, the change in the SLBT runtime with the eyes open and the dominant leg was significantly higher in the robotic surgery protocol. With closed eyes and the non-dominant leg, the change in the SLBT runtime was significantly higher in the laparoscopy surgery protocol. There were no significant differences in SLBT runtime between the non-dominant leg with eyes closed and the eyes open with dominant leg scenario.

When comparing the changes in the assistant surgeon after the robotic and laparoscopy surgery protocol, there were significantly higher changes in six Surgeons' PROMS ($POMS_{Index}$, $POMS_{Fatigue}$, $POMS_{Vigour}$, $POMS_{Friendliness}$, $POMS_{Anxiety}$, and QPFS) after the laparoscopy surgery protocol, while in two Surgeons' PROMS, the changes were significantly higher after the robotic surgery protocol

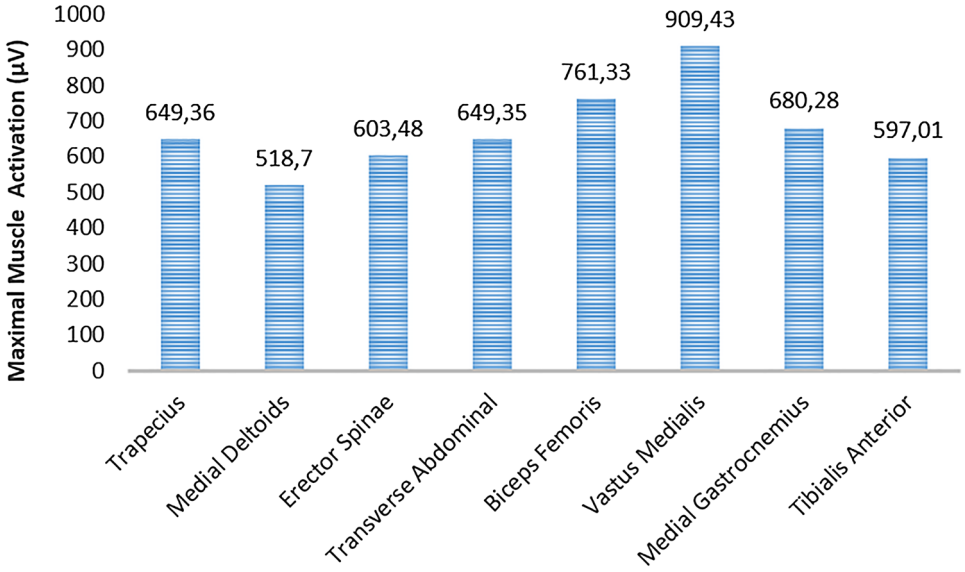

Fig. 2 MVC of each muscle analysed during the surgical intervention

Table 1 Change in fatigue, "intra-profile" (main surgeon and assistant surgeon) and "inter-profile," the surgeon after surgical interventions with chief surgeon surgery protocol

	Robotic			Laparoscopic			Dif profile post-inter.
	Pre	Post	Dif.	Pre	Post	Dif.	
PRO							
POMS							
POMS Index	7.00	40.50	33.50***	4.50	26.00	21.50***	12.00***
Anger	0.50	6.50	6.00**	0.00	2.50	2.50***	3.5***
Depression	0.00	1.50	1.50*	0.00	1.00	1.00**	0.5**
Fatigue	0.50	3.00	2.50*	0.50	2.50	2.00**	0.5**
Vigor	2.50	12.00	9.50***	1.50	7.50	6.00**	3.5*
Friendliness	2.50	11.50	9.00***	1.50	8.00	6.50***	2.5**
Anxiety	0.50	3.50	3.00***	1.00	2.50	1.50*	1.5***
Confusion	0.50	2.50	2.00**	0.00	2.00	2.00**	0
QPFS	16.00	68.00	52.00***	15.50	68.50	53.00**	−1
VAS	21.80	72.60	50.80***	20.90	64.00	43.10**	7.7***
Functional test							
Handgrip (kg)							
Maximum left	39.50	35.00	−4.50***	39.00	30.50	8.50***	4.5**
Maximum right	40.50	36.50	−4.00***	42.00	32.50	9.50***	5.5***
Single-leg balance test							
Dominant leg							
Opened eyes							
Time (seg)	30.00	16.25	−14.75***	30.00	19.34	−11.75***	3.0**
Displacement (°)							
Flex/Ext	2.24	21.09	18.85***	4.84	23.92	19.08**	−0.23
Incl. L/R	9.58	24.55	14.97***	9.49	21.26	11.77**	3.2***
Rot L/R	9.87	35.24	25.37***	9.86	28.67	18.81*	6.56**
Velocity (°/s)							
Flex/Ext	40.67	46.18	5.51*	41.62	43.23	1.61	3.9**
Incl. L/R	23.75	30.48	6.73*	23.67	25.00	1.33	5.4***
Rot L/R	51.29	58.04	6.75**	52.49	56.44	3.95	2.8*
Closed eyes							
Time (seg)	30.00	9.28	−20.72***	30.00	10.57	−19.43***	1.29
Displacement (°)							
Flex/Ext	9.51	48.48	38.97***	9.29	48.97	39.68***	−0.71
Incl. L/R	14.01	53.09	39.08***	14.50	44.54	30.04***	9.04**
Rot L/R	15.29	57.73	42.44***	15.68	46.89	31.21***	11.23***
Velocity (°/s)							
Flex/Ext	42.09	46.15	4.06*	42.66	42.28	−0.38	4.44***
Incl. L/R	29.67	30.79	1.12	29.63	32.81	3.18*	−2.06*
Rot L/R	56.38	55.07	−1.31	57.45	57.12	−0.33	−0.98
Non-dominant leg							
Opened eyes							
Time (seg)	30.00	16.47	−13.53***	30.00	16.34	−13.66***	−0.13
Displacement (°)							
Flex/Ext	5.84	24.87	19.03***	6.06	20.12	14.06**	4.97***
Incl. L/R	11.94	26.53	14.59***	11.78	22.45	10.67*	3.92**
Rot L/R.	12.33	40.21	27.88***	12.65	33.21	20.56***	7.32***
Velocity (°/s)							
Flex/Ext	35.68	43. 58	7.90**	35.31	44.78	9.47**	−1.57*
Incl. L/R	36.39	44.32	7.93*	37.23	31.74	−5.49*	2.44**

Table 1 continued

	Robotic			Laparoscopic			Dif profile post-inter.
	Pre	Post	Dif.	Pre	Post	Dif.	
Rot L/R	52.60	56.37	3.77	53.59	59.62	6.03**	−2.26**
Closed eyes							
Time (seg)	26.45	4.84	−21.61***	25.97	6.77	−19.20***	−2.41*
Displacement (°)							
Flex/Ext	14.32	59.24	44.92***	13.87	49.09	35.22***	9.7**
Incl. L/R	19.21	63.44	44.23***	19.68	53.70	34.02**	10.21***
Rot L/R	20.74	63.18	42.44***	21.48	52.44	30.96***	11.48***
Velocity (°/s)							
Flex/Ext	43.55	44.87	1.32	45.06	42.17	−2.89*	−4.21**
Incl. L/R	27.55	34.15	6.60**	28.41	40.29	11.88**	−5.28***
Rot L/R	58.24	60.44	2.20	60.27	63.28	3.01**	−0.81

Pre pre-intervention, *Post* post-intervention, *Dif* difference, *Dif Profile Post-Inter* differences between both surgeon profiles (chief and assistant surgeon), *PRO* Patient-Reported Outcome, *QPFS* Quick Piper Fatigue Scale, *POMS* Profile Mood States, *VAS* Visual Analogue Scale, *Flex/Ext* flexion–extension, *Incl. L/R* left–right inclination, *Rot L/R* left–right rotation

Significance (p): * ≤0.05; ** ≤0.005; *** ≤0.001

($POMS_{Confusion}$ and VAS). There were no significant differences in $POMS_{Anger}$ and $POMS_{Depression}$ between the two protocols (Table 2). Nevertheless, in the assistant surgeon, the changes observed in functional tests after the laparoscopy surgery protocol were not significant for grip strength in either hand, compared with robotic surgery protocol. Finally, in the SLBT, significant differences in the changes produced were observed in only 5 of the 24 kinematic variables recorded after the laparoscopy protocol (Table 2), with the highest difference identified (10.51°) in the left/right lateral inclination measured during SLBT with the eyes closed. The change in the SLBT runtime with the dominant leg was significantly higher after the laparoscopy surgery protocol; no significant differences were observed between protocols in terms of the SLBT runtime with the non-dominant leg.

Inter-profile differences after the same protocol

Comparison of the inter-profile differences in variables measured pre- and post-laparoscopic intervention revealed significant differences in all Surgeons' PROMS and OCOM variables after laparoscopic surgery, with the exception of velocities measured during the SLBT in both the chief and assistant surgeon profiles (Table 4). In the post-intervention inter-profile comparison, there were significant differences in all three Surgeons' PROMS variables measured (POMS 3.1 %; QPFS 8 %; 12.5 % VAS) after laparoscopic surgery protocol, although the magnitude of the inter-profile difference was lower than in the post-robotic surgery protocol (Table 4), which showed significant differences of 26 % (POMS), 14.50 (QPFS) and 16.25 % (VAS) (Table 3). As regards the OCOM variables, the handgrip was significantly weaker in the chief than in the assistant surgeon profile after both protocols, but the changes were higher after laparoscopy [4.00 kg (right hand) and 5.50 kg (left hand)] (Table 4), compared with the robotic surgery protocol (0.50 kg—both hands). Nevertheless, the behaviour of the SLBT kinematic variables measured after both protocols differed between roles. Significant differences were observed between the chief and assistant surgeon in 25 of the 28 kinematic variables measured after the robotic surgery protocol (Table 3). After the laparoscopy surgery protocol, a significant difference was observed between the two profiles in 9 of the 28 variables analysed.

Variables measured during surgery

Tables 5 and 6 show the kinematic variables recorded during surgical interventions in terms of surgeon profile (assistant and chief) per protocol (robotic and laparoscopy). Table 5 shows that there were significant differences in the kinematic variables between surgeon profiles during the robotic protocol, whereas there were no significant differences in the laparoscopy protocol. Similarly, there were significant differences in kinematic variables of the chief surgeon between the two protocols (Table 6). However, there were no significant differences between protocols in terms of the kinematic records of the assistant surgeon (Table 6).

Table 2 Change in fatigue, "intra-profile" (main surgeon and assistant surgeon) and "inter-profile," the surgeon after surgical interventions with assistant surgeon surgery protocol

	Robotic			Laparoscopic			Dif profile post-inter.
	Pre	Post	Dif.	Pre	Post	Dif.	
PRO							
POMS							
POMS Index	6.00	13.50	7.50***	3.50	17.00	13.50***	−6***
Anger	0.00	2.00	2.00**	0.50	2.50	2.00**	0
Depression	0.00	0.00	0.00**	0.00	0.50	0.50*	−0.5
Fatigue	0.00	1.00	1.00*	0.00	1.50	1.50*	−0.5*
Vigor	2.50	4.50	2.00**	1.50	5.50	4.00**	−2**
Friendliness	2.50	5.00	2.50**	1.50	5.50	4.00**	−1.5**
Anxiety	0.50	1.00	0.50*	0.00	1.50	1.50*	−1**
Confusion	0.50	1.00	0.50*	0.00	0.00	0.00**	0.5*
QPFS	16.00	53.50	37.50***	16.50	57.50	41.00***	−3.5***
VAS	20.35	54.90	34.55***	21.20	51.80	30.60**	3.95***
Functional test							
Handgrip (kg)							
Maximum left	39.50	35.50	4.00**	39.50	35.00	4.50**	−0.5
Maximum right	41.50	38.00	3.50**	41.50	37.50	4.00***	−0.5
Single-leg balance test							
Dominant leg							
Opened eyes							
Time (seg)	30.00	23.49	−7.11***	30.00	21.93	−8.07***	0.96**
Displacement (°)							
Flex/Ext	2.28	17.24	14.96***	3.08	17.33	14.25***	0.71
Incl. L/R	9.68	20.11	10.43***	9.68	20.29	10.61***	−0.18
Rot L/R.	9.67	29.35	19.68***	9.67	29.37	19.70**	−0.02
Velocity (°/s)							
Flex/Ext	41.89	44.47	2.58	41.89	43.45	1.56	1.02
Incl. L/R	23.51	25.15	1.64	23.51	25.23	1.72	−0.08
Rot L/R	55.57	57.80	2.23	55.57	56.47	0.90	1.33
Closed eyes							
Time (seg)	30.00	12.87	−16.53***	30.00	11.69	−18.31***	1.78*
Displacement (°)							
Flex/Ext	9.41	38.18	28.77***	9.41	39.10	29.69***	−0.92
Incl. L/R	14.43	54.03	39.60***	14.43	43.52	29.09**	10.51***
Rot L/R	15.35	47.36	32.01***	15.35	46.18	30.83**	1.18
Velocity (°/s)							
Flex/Ext	42.26	43.88	1.62	42.26	43.30	1.04	0.58
Incl. L/R	28.56	31.67	3.11*	30.56	31.71	1.15	1.96*
Rot L/R	54.97	56.75	1.78	54.97	55.69	0.72	1.06
Non-dominant leg							
Opened eyes							
Time (seg)	30.00	20.55	−9.14***	30.00	19.86	−10.14***	1
Displacement (°)							
Flex/Ext	5.83	21.89	16.06***	5.83	20.15	14.32**	1.74*
Incl. L/R	11.65	21.74	10.09***	11.65	22.03	10.38***	−0.29
Rot L/R	12.30	32.92	20.62***	12.30	32.09	19.79**	0.83
Velocity (°/s)							
Flex/Ext	39.74	42.75	3.01	39.74	43.19	3.45**	−0.44
Incl. L/R	26.38	32.91	6.53*	26.38	32.17	5.79**	0.74

Table 2 continued

	Robotic			Laparoscopic			Dif profile post-inter.
	Pre	Post	Dif.	Pre	Post	Dif.	
Rot L/R	56.55	59.21	2.66	56.55	58.12	1.57	1.09
Closed eyes							
Time (seg)	26.60	7.71	−18.89***	24.61	7.17	−17.44**	−1.45
Displacement (°)							
Flex/Ext	14.83	46.48	31.65***	14.83	47.98	33.15***	−1.5*
Incl. L/R	19.36	54.01	34.65***	19.36	52.71	33.35***	1.3
Rot L/R	22.16	54.36	32.20***	22.16	52.50	30.34***	1.86**
Velocity (°/s)							
Flex/Ext	40.50	45.04	4.54*	40.50	43.53	3.03**	1.51*
Incl. L/R	29.48	40.55	11.07**	29.48	39.32	9.84**	1.23
Rot L/R	56.19	63.24	7.05*	56.19	61.11	4.92**	2.13**

Pre pre-intervention, *Post* post-intervention, *Dif* difference, *Dif Profile Post-Inter* differences between both surgeon profiles (chief and assistant surgeon), *PRO* Patient-Reported Outcome, *QPFS* Quick Piper Fatigue Scale, *POMS* Profile Mood States; *VAS* Visual Analogue Scale, *Flex/Ext* flexion–extension, *Incl L/R* left–right inclination, *Rot L/R* left–right rotation

Significance (p): * $\leq$0.05; ** $\leq$0.005; *** $\leq$0.001

Figure 3 presents the mean activation values of each of the eight muscles analysed for the surgeon profile and during either robotic or laparoscopic protocols. It is apparent that median frequency muscle activation only exceeded 25 % MVC (27.33 % MVC) in the erector spinae, while the MDF from other muscles ranged from 9.87 % MVC (biceps femoralis–chief surgeon–robotic protocol) to 19.40 % MVC (trapecious–chief surgeon–robotic protocol).

Finally, Table 7 presents the MDFslope *I*, presented in absolute (Hz/min) and relative values (%/min). Figure 4 shows a graphic example of the calculation of the MDFslope. Both Fig. 4 and Table 7 shows that there was no clearly defined trend in the positive and negative MDFslope values, or within the muscle in different situations (within the same surgeon profile and surgical intervention).

Discussion

The aim of this study was to analyse the onset and evolution of fatigue experienced by a surgeon (chief and assistant) before, during and after surgical interventions using robotic (Da Vinci®) and laparoscopic surgery protocols. The results showed that there were significant differences in all outcome variables analysed before and after surgery; however, the magnitude of the differences was different depending on the profile and the surgical procedure performed (Tables 1, 2). In this sense, there were significant differences in the kinematic variables recorded during surgery (Tables 3, 4). Based on these results, the aim of the study was achieved and the hypothesis was confirmed.

Pre- and post-intervention variables

Intra-profile comparison

When we analysed the results of the Surgeons' PROMS and functional test pre- and post-intervention, we observed significant differences in all outcome variables (Tables 1, 2); however, the changes differed depending on the surgical profile and intervention. Thus, the Surgeons' PROMS variables experienced a greater change in the chief surgeon profile (POMS 8.27–12.88 %; QPFS 28.67–36.67 %; VAS 50.80–53.00 %) compared to the assistant surgeon profile (POMS 2.88–5.19 %; QPFS 25–27.33 %; VAS 30.60–34.55 %) (Tables 1, 2). These differences in the subjective aspect of fatigue may be attributed to the decision-making process and the execution of the technique during surgery, which were two aspects that increased fatigue [29, 30].

After performing the surgery, significant changes (from pre- to post-intervention) were observed in all OCOM variables. In the handgrip test, the chief surgeon experienced a greater change following laparoscopy (right 9.50 kg and left 8.50 kg) with respect to the other scenarios, in which the strength decreased by 3.5–4.5 kg (Tables 1, 2). This difference could be due to the fact that during robotic surgery, the surgeon experienced little or no

Table 3 Change in fatigue, "intra-profile" (main surgeon and assistant surgeon) and "inter-profile," the surgeon after surgical interventions with robotic surgery protocol

	Chief surgeon			Assistant surgeon			Dif profile post-inter.
	Pre	Post	Dif.	Pre	Post	Dif.	
PRO							
POMS							
POMS Index	7.00	40.50	33.50***	6.00	13.50	7.50***	−26.00***
Anger	0.50	6.50	6.00**	0.00	2.00	2.00**	−4.00**
Depression	0.00	1.50	1.50*	0.00	0.00	0.00**	−1.50**
Fatigue	0.50	3.00	2.50*	0.00	1.00	1.00*	−1.50***
Vigor	2.50	12.00	9.50***	2.50	4.50	2.00**	−7.50**
Friendliness	2.50	11.50	9.00***	2.50	5.00	2.50**	−6.50**
Anxiety	0.50	3.50	3.00***	0.50	1.00	0.50*	−2.50**
Confusion	0.50	2.50	2.00**	0.50	1.00	0.50*	−1.50*
QPFS	16.00	68.00	52.00***	16.00	53.50	37.50***	−14.50**
VAS	21.80	72.60	50.80***	20.35	54.90	34.55***	−16.25***
Functional test							
Handgrip (kg)							
Maximum left	39.50	35.00	−4.50***	39.50	35.50	4.00**	−0.50**
Maximum right	40.50	36.50	−4.00***	41.50	38.00	3.50**	−0.50**
Single-leg balance test							
Dominant leg							
Opened eyes							
Time (seg)	30.00	16.25	−14.75***	30.00	23.49	−7.11***	7.64***
Displacement (°)							
Flex/Ext	2.24	21.09	18.85***	2.28	17.24	14.96***	−3.89*
Incl. L/R	9.58	24.55	14.97***	9.68	20.11	10.43***	−4.54**
Rot L/R	9.87	35.24	25.37***	9.67	29.35	19.68***	−5.69**
Velocity (°/s)							
Flex/Ext	40.67	46.18	5.51*	41.89	44.47	2.58	−2.93*
Incl. L/R	23.75	30.48	6.73*	23.51	25.15	1.64	−5.09**
Rot L/R	51.29	58.04	6.75**	55.57	57.80	2.23	−4.52*
Closed eyes							
Time (seg)	30.00	9.28	−20.72***	30.00	12.87	−16.53***	4.19**
Displacement (°)							
Flex/Ext	9.51	48.48	38.97***	9.41	38.18	28.77***	−10.20***
Incl. L/R	14.01	53.09	39.08***	14.43	54.03	39.60***	0.52
Rot L/R	15.29	57.73	42.44***	15.35	47.36	32.01***	−10.43**
Velocity (°/s)							
Flex/Ext	42.09	46.15	4.06*	42.26	43.88	1.62	−2.44*
Incl. L/R	29.67	30.79	1.12	28.56	31.67	3.11*	2.99*
Rot L/R	56.38	55.07	−1.31	54.97	56.75	1.78	3.09**
Non-dominant leg							
Opened eyes							
Time (seg)	30.00	16.47	−13.53***	30.00	20.55	−9.14***	4.39***
Displacement (°)							
Flex/Ext	5.84	24.87	19.03***	5.83	21.89	16.06***	−2.97*
Incl. L/R	11.94	26.53	14.59***	11.65	21.74	10.09***	−4.50*
Rot L/R	12.33	40.21	27.88***	12.30	32.92	20.62***	−7.26*
Velocity (°/s)							
Flex/Ext	35.68	43. 58	7.90**	39.74	42.75	3.01	−4.89*
Incl. L/R	36.39	44.32	7.93*	26.38	32.91	6.53*	1.40

Table 3 continued

	Chief surgeon			Assistant surgeon			Dif profile post-inter.
	Pre	Post	Dif.	Pre	Post	Dif.	
Rot L/R	52.60	56.37	3.77	56.55	59.21	2.66	−1.11
Closed eyes							
Time (seg)	26.45	4.84	−21.61***	26.60	7.71	−18.89***	2.72*
Displacement (°)							
Flex/Ext	14.32	59.24	44.92***	14.83	46.48	31.65***	−13.27***
Incl. L/R	19.21	63.44	44.23***	19.36	54.01	34.65***	−9.58**
Rot L/R	20.74	63.18	42.44***	22.16	54.36	32.20***	−10.24***
Velocity (°/s)							
Flex/Ext	43.55	44.87	1.32	40.50	45.04	4.54*	3.22*
Incl. L/R	27.55	34.15	6.60**	29.48	40.55	11.07**	4.47*
Rot L/R	58.24	60.44	2.20	56.19	63.24	7.05*	4.85*

Pre pre-intervention, *Post* post-intervention, *Dif* difference, *Dif Profile Post-Inter* differences between both surgeon profiles (chief and assistant surgeon), *PRO* Patient-Reported Outcome, *QPFS* Quick Piper Fatigue Scale, *POMS* Profile Mood States, *VAS* Visual Analogue Scale, *Flex/Ext* flexion–extension, *Incl L/R* left–right inclination, *Rot L/R* left–right rotation

Significance (p): * ≤0.05; ** ≤0.005; *** ≤0.001

resistance to the use of the instruments. This feature is one of the great advantages of robotic surgery compared to laparoscopy [5].

On the other hand, when analysing the results of the SLBT in tests with the dominant leg with the eyes open and closed, the changes experienced by the chief surgeon in the robotic condition were greater than those experienced in the other scenarios. Differences were also significant when comparing the data for the assistant following the robotic and the laparoscopic protocols (Tables 5, 6, 7). Both the results of the handgrip and the SLBT were contrary to those found in a previous study [20] in which, after comparing the fatigue of surgeons in robotic and laparoscopy protocols, the authors found no significant differences between the two protocols. The differences between studies may reflect differences in the duration of surgery. In the present study, the mean duration of surgery was 218.46 (±14.55) min, while the surgeries analysed in the previously mentioned study [20] had a duration of 89 and 114 min for the laparoscopic and robotic protocols, respectively. This increased operating time coupled with the reduction in ergonomics identified elsewhere [5] as one of the principle issues in robotic surgery might explain the differences in the SLBT results observed in the present study.

Inter-profile comparison

Comparison of the differences between the chief surgeon after laparoscopy and robotic surgery protocol revealed significant differences in most of the Surgeons' PROMS variables. These results indicate that fatigue perceived by the surgeon after the robotic surgery protocol was significantly higher compared to the laparoscopy surgery protocol (Table 1). The VAS variable deserves special attention because it was a question that the surgeon could interpret in a more open way, integrating concepts and feelings that perhaps were not included in the POMS and QPFS variables. The difference in the change experienced by the chief surgeon after the robotic surgery protocol was 7.5 %. Likewise, the change in the 15 kinematic variables recorded during SLBT during the four conditions was significantly higher after the robotic protocol than after the laparoscopy protocol (Table 1). During the robotic surgery protocol, the chief surgeon must maintain the same position throughout surgery and cannot modify his position freely. This could increase the physical fatigue and worsen the mental fatigue experienced by the chief surgeon because of the constant decisions made during surgery, which would explain the difference in fatigue perceived after laparoscopy and robotic surgery.

Regarding the change in mental fatigue perceived by the assistant surgeon, there were significant differences in 6 of the 10 Surgeons' PROMS variables analysed. One reason for this could be that during robotic surgery, the main function of the assistant surgeon is to exchange the "arms" of the robotic system according to the needs of the chief surgeon, while during laparoscopy, the assistant surgeon must concentrate on giving the chief surgeon the best image through the camera that he holds. These differences

Table 4 Change in fatigue, "intra-profile" (main surgeon and assistant surgeon) and "inter-profile," the surgeon after surgical interventions with LAPAROSCOPY surgery protocol

	Chief surgeon			Assistant surgeon			Dif profile post-inter.
	Pre	Post	Dif.	Pre	Post	Dif.	
PRO							
POMS							
POMS Index	4.50	26.00	21.50***	3.50	17.00	13.50***	8.00**
Anger	0.00	2.50	2.50***	0.50	2.50	2.00**	0.50
Depression	0.00	1.00	1.00**	0.00	0.50	0.50*	0.50
Fatigue	0.50	2.50	2.00**	0.00	1.50	1.50*	0.50
Vigor	1.50	7.50	6.00**	1.50	5.50	4.00**	2.00*
Friendliness	1.50	8.00	6.50***	1.50	5.50	4.00**	2.50**
Anxiety	1.00	2.50	1.50*	0.00	1.50	1.50*	0.00
Confusion	0.00	2.00	2.00**	0.00	0.00	0.00**	2.00*
QPFS	15.50	68.50	53.00**	16.50	57.50	41.00***	12.00**
VAS	20.90	64.00	43.10**	21.20	51.80	30.60**	12.50**
Functional test							
Handgrip (kg)							
Maximum Left	39.00	30.50	8.50***	39.50	35.00	4.50**	4.00***
Maximum Right	42.00	32.50	9.50***	41.50	37.50	4.00***	5.50***
Single-leg balance test							
Dominant leg							
Opened Eyes							
Time (seg)	30.00	19.34	−11.75***	30.00	21.93	−8.07***	3.68**
Displacement (°)							
Flex/Ext	4.84	23.92	19.08**	3.08	17.33	14.25***	−4.83*
Incl. L/R	9.49	21.26	11.77**	9.68	20.29	10.61***	−1.16
Rot L/R	9.86	28.67	18.81*	9.67	29.37	19.70**	0.89
Velocity (°/s)							
Flex/Ext	41.62	43.23	1.61	41.89	43.45	1.56	−0.05
Incl. L/R	23.67	25.00	1.33	23.51	25.23	1.72	0.39
Rot L/R	52.49	56.44	3.95	55.57	56.47	0.90	−3.05*
Closed eyes							
Time (seg)	30.00	10.57	−19.43***	30.00	11.69	−18.31***	1.12
Displacement (°)							
Flex/Ext	9.29	48.97	39.68***	9.41	39.10	29.69***	10.01***
Incl. L/R	14.50	44.54	30.04***	14.43	43.52	29.09**	−0.95
Rot L/R	15.68	46.89	31.21***	15.35	46.18	30.83**	−0.38
Velocity (°/s)							
Flex/Ext	42.66	42.28	−0.38	42.26	43.30	1.04	1.42
Incl. L/R	29.63	32.81	3.18*	30.56	31.71	1.15	−2.03
Rot L/R	57.45	57.12	−0.33	54.97	55.69	0.72	1.05
Non-dominant leg							
Opened eyes							
Time (seg)	30.00	16.34	−13.66***	30.00	19.86	−10.14***	3.52*
Displacement (°)							
Flex/Ext	6.06	20.12	14.06**	5.83	20.15	14.32**	0.26
Incl. L/R	11.78	22.45	10.67*	11.65	22.03	10.38***	−0.29
Rot L/R	12.65	33.21	20.56***	12.30	32.09	19.79**	−0.77
Velocity (°/s)							
Flex/Ext	35.31	44.78	9.47**	39.74	43.19	3.45**	−6.02**
Incl. L/R	37.23	31.74	−5.49*	26.38	32.17	5.79**	11.28*

Table 4 continued

	Chief surgeon			Assistant surgeon			Dif profile post-inter.
	Pre	Post	Dif.	Pre	Post	Dif.	
Rot L/R	53.59	59.62	6.03**	56.55	58.12	1.57	−4.46*
Closed eyes							
Time (seg)	25.97	6.77	−19.20***	24.61	7.17	−17.44**	1.76
Displacement (°)							
Flex/Ext	13.87	49.09	35.22***	14.83	47.98	33.15***	−2.07
Incl. L/R	19.68	53.70	34.02**	19.36	52.71	33.35***	−0.67
Rot L/R.	21.48	52.44	30.96***	22.16	52.50	30.34***	−0.62
Velocity (°/s)							
Flex/Ext	45.06	42.17	−2.89*	40.50	43.53	3.03**	5.92*
Incl. L/R	28.41	40.29	11.88**	29.48	39.32	9.84**	−2.04
Rot L/R	60.27	63.28	3.01**	56.19	61.11	4.92**	1.91

Pre pre-intervention, *Post* post-intervention, *Dif* difference, *Dif Profile Post-Inter* differences between both surgeon profiles (chief and assistant surgeon), *PRO* Patient-Reported Outcome, *QPFS* Quick Piper Fatigue Scale, *POMS* Profile Mood States, *VAS* Visual Analogue Scale, *Flex/Ext* flexion–extension, *Incl L/R* left–right inclination, *Rot L/R* left–right rotation

Significance (p): * $\leq$0.05; ** $\leq$0.005; *** $\leq$0.001

Table 5 Median values of muscle activation frequency slope for each of the eight muscles analysed for both profiles surgeon (and assistant chief) and the two protocols of surgery (robotic–laparoscopy)

	Robotic				Laparoscopy			
	Chief surgeon		Assistant surgeon		Chief surgeon		Assistant surgeon	
	Hz/min	%/min	Hz/min	%/min	Hz/min	%/min	Hz/min	%/min
Trapecius	1.7554	0.0029	−6.4463	−0.0093	−2.5318	−0.0042	5.9103	0.0111
Medial deltoids	−1.4917	−0.0025	6.4848	0.0094	−1.9382	−0.0032	−5.5662	−0.0105
Erector spinae	3.7758	0.0063	2.0991	0.0035	−6.4352	−0.0107	−3.2844	−0.0054
Transverse abdominal	−1.2337	−0.0020	5.1128	−0.0074	1.5192	0.0025	4.6192	0.0087
Biceps femoris	1.6331	0.0027	7.0995	0.0103	−2.1219	−0.0035	−3.0469	−0.0058
Vastus medialis	−1.9218	−0.0032	7.4441	0.0108	2.5947	0.0043	3.5090	0.0061
Medial gastrocnemius	1.4889	0.0025	6.4250	0.0093	−1.9710	−0.0033	5.6186	0.0106
Tibialis anterior	−1.5606	−0.0026	−5.6117	−0.0081	−2.2045	−0.0037	5.2974	0.0100

in changes in Surgeons' PROMS variables were not reflected in the variables of the functional tests: No significant differences were observed in the handgrip test, while in the balance tests, significant differences were only observed in 9 of the 28 variables analysed (Table 2). One possible explanation for these more equal results (especially in functional variables) could be found in the higher mobility that the assistant surgeon had relative to the chief surgeon during both protocols.

Variables measured during surgery

In the present study, we considered a surgeon's mobility and his muscle activity during surgical interventions. Based on the records collected during surgery, the chief surgeon demonstrated significantly lower mobility in the robotic protocol than in the other scenarios (Tables 3, 5), while no significant differences were observed among the other profiles (Tables 4, 5). The increased fatigue caused by the lack of mobility may be one reason why the lack of ergonomics had been identified as a problem in robotic surgery. Previous studies have shown that maintaining the same non-ergonomic posture causes (in the short term) the early onset of fatigue and (in the long term) the development of musculoskeletal problems, a high percentage of which are debilitating [31].

On the contrary, none of the muscles studied during surgery exceeded an activation of 25 % MVC, except for

Table 6 Record kinematic variables made during surgery and differences between the two protocols of intervention (robotic or laparoscopy) profile for the same surgeon (chief or assistant surgeon)

	Chief surgeon			Assistant surgeon		
	Robotic	Lapar.	Dif.	Robotic	Lapar.	Dif.
Displacement (°)						
Flexion/extension	8.72 (±19.30)	41.09 (±8.66)	32.37***	38.33 (±6.35)	39.84 (±6.77)	1.51
Left/right inclination	14.59 (±34.80)	38.48 (±7.39)	23.89***	43.37 (±5.73)	41.90 (±6.09)	−1.47
Left/right rotation	9.65 (±23.76)	43.48 (±6.97)	33.83***	40.39 (±7.02)	41.31 (±6.73)	0.92
Velocity (°/s)						
Flexion/extension	2.37 (±46.91)	19.76 (±10.03)	17.39***	19.27 (±8.22)	18.91 (±8.44)	−0.36
Left/right Inclination	4.29 (±39.57)	18.43 (±8.59)	14.14***	17.27 (±10.66)	16.55 (±9.88)	−0.72
Left/right Rotation	2.69 (±37.33)	17.68 (±9.17)	14.99***	17.08 (±9.09)	18.49 (±10.38)	1.41

Lapar laparoscopy; *Dif* difference

Significance (*p*): * ≤0.05; ** ≤0.005; *** ≤0.001

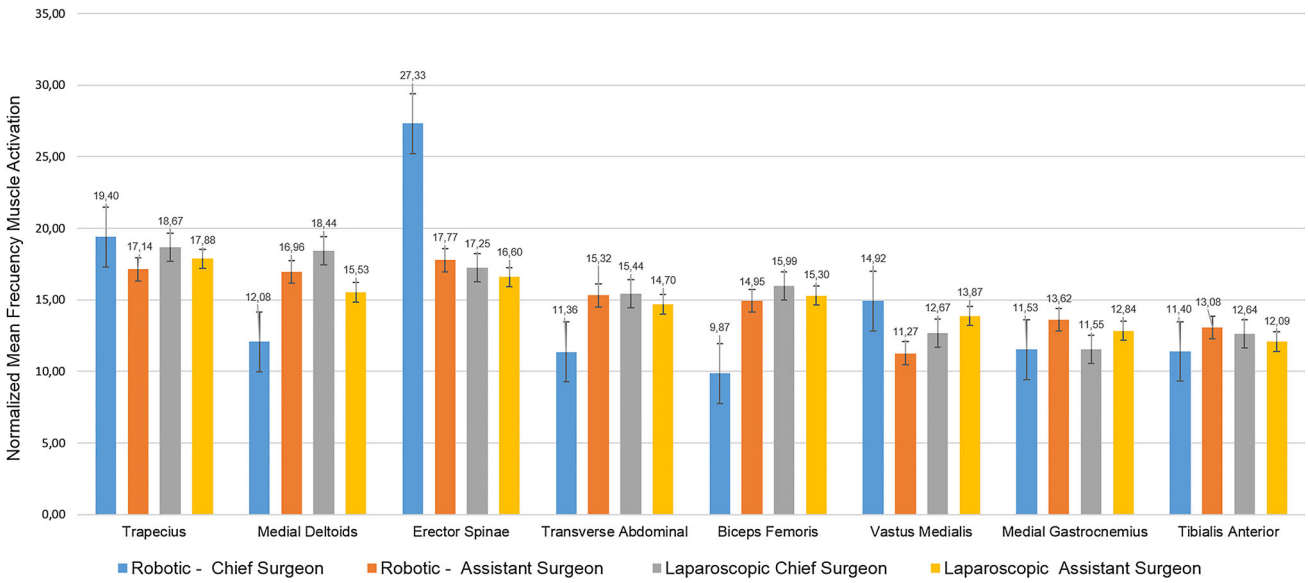

Fig. 3 Mean frequency muscle activation recorded during the surgical intervention for each muscle analysed

the erector spinae, in which the activation reached an average of 27.33 % MVC (chief surgeon–robotic protocol) (Fig. 3). Thus, muscle activation did not reach the minimum level of activation to cause muscle fatigue [32]. This trend was further confirmed by the MDFslope, which lay between −0.0107 %/min (erector spinae–chief surgeon–laparoscopy) and 0.0111 %/min (trapezius–assistant surgeon–laparoscopy). The rate of change in the MDFslope was minimal and, in many muscles, was not negative. Thus, although muscle activity was comparable between profiles and protocols, the results of the test indicated lower post-intervention fatigue in the chief surgeon–robotic protocol. These results may be attributed to the position of the chief surgeon during the robotic protocol, as he was seated with minimal capacity for movement. Thus, one could argue that muscle activity was required to maintain the position for a long time, while in the other three profiles, the surgeon had a greater range of motion. Previous studies have shown that muscle contractions during dynamic activity promote activation recovery of individual muscle

Table 7 Median values of muscle activation frequency slope for each of the eight muscles analysed for both profiles surgeon (and assistant chief) and the two protocols of surgery (robotic–laparoscopy)

	Robotic				Laparoscopy			
	Chief surgeon		Assistant surgeon		Chief surgeon		Assistant surgeon	
	Hz/min	%/min	Hz/min	%/min	Hz/min	%/min	Hz/min	%/min
Trapecius	1.7554	0.0029	−6.4463	−0.0093	−2.5318	−0.0042	5.9103	0.0111
Medial deltoids	−1.4917	−0.0025	6.4848	0.0094	−1.9382	−0.0032	−5.5662	−0.0105
Erector spinae	3.7758	0.0063	2.0991	0.0035	−6.4352	−0.0107	−3.2844	−0.0054
Transverse abdominal	−1.2337	−0.0020	5.1128	−0.0074	1.5192	0.0025	4.6192	0.0087
Biceps femoris	1.6331	0.0027	7.0995	0.0103	−2.1219	−0.0035	−3.0469	−0.0058
Vastus medialis	−1.9218	−0.0032	7.4441	0.0108	2.5947	0.0043	3.5090	0.0061
Medial gastrocnemius	1.4889	0.0025	6.4250	0.0093	−1.9710	−0.0033	5.6186	0.0106
Tibialis anterior	−1.5606	−0.0026	−5.6117	−0.0081	−2.2045	−0.0037	5.2974	0.0100

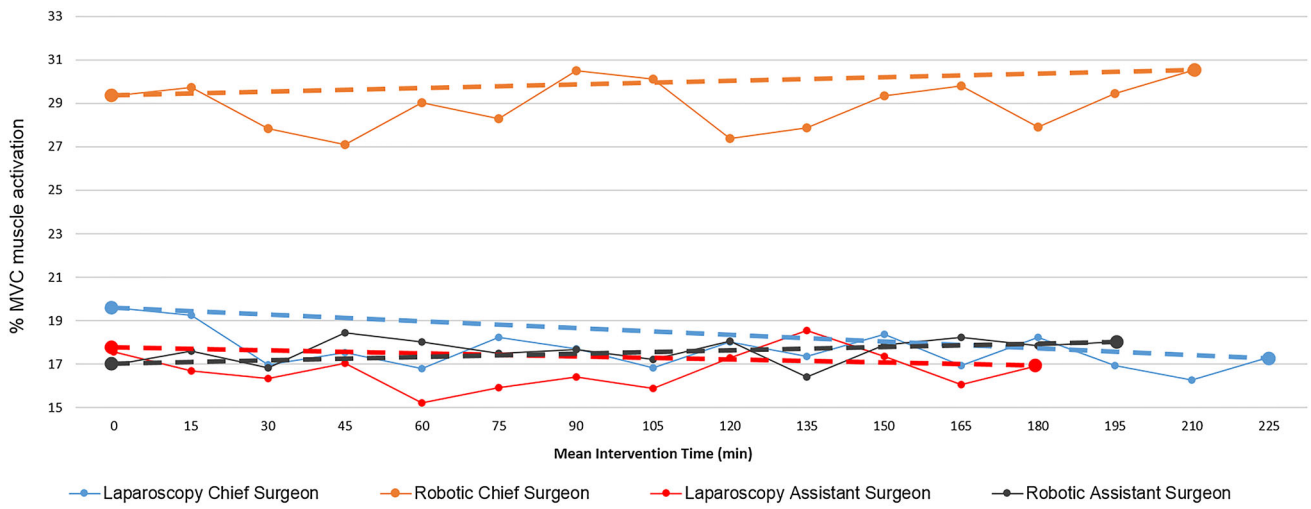

Fig. 4 % MVC muscle activation (erector spinae) for two surgeon profiles (chief and assistant surgeon) and for two intervention protocols (Da Vinci and Laparoscopy) with the fatigue tendency (*dashed line*)

fibres, causing a better distribution of the stimulus along the entire muscle [33]. Conversely, maintaining the same position during surgery caused increased fatigue as well as reduced motor control post-intervention [34]. In fact, one study suggested the need for micropauses (every 30 min) in prolonged surgical interventions, which would help to decrease fatigue and its deleterious effects [35].

Implication of the research

According to the results of this study, fatigue related to the development of robot–surgeon interaction can influence the results of surgery. Thus, both the most experienced surgeons and medical residents who join the different hospital services should plan strategies to avoid or limit the onset of fatigue during surgery, such as, for example, improved physical condition.

Limits and strengths

To our knowledge, this is the first study to analyse the mobility and muscular activity of a surgeon performing long-duration surgery (more than 2 h). It is also the first to compare the results depending on the surgeon profile (assistant and chief surgeon) and intervention protocol (laparoscopic and robotic). However, this study also has some limitations. For example, the study was conducted on the same surgeon (over 15 years of experience in general surgery and surgery on the digestive tract) in the two analysed profiles (chief and assistant) for the two protocols

(robotic and laparoscopic). Future studies should increase the number of participating surgeons with the same profile as the surgeon analysed to consolidate the results observed in the present study. In addition, future work could focus on other types of surgery and other profiles: both personal (different genders and ages) and professional (less experience or from other medical specialties).

Conclusion

The main conclusion that can be reached after analysing the results of the present study is that long-duration surgery causes increased fatigue in the surgeon, resulting in increased fatigue perceived by the surgeon and decreased handgrip strength and motor control. However, the magnitude of fatigue is different when two different surgeon profiles (assistant and chief) and intervention protocols (robotic and laparoscopic) are compared. The profile of the surgeon who experienced the most fatigue was the chief surgeon in the robotic protocol. This increased fatigue can be attributed to two main factors: the surgical decision-making process and the static position that must be maintained for long periods, which was demonstrated by the kinematic record and the muscle activity during surgery.

Compliance with ethical standards

Disclosure All authors, Manuel González-Sánchez, Ivan González-Poveda, Santiago Mera-Velasco and Antonio I. Cuesta-Vargas, declare that they have no conflicts of interest or financial ties to disclose.

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